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SECTION 02

GROUP BIII

PROJECT:

**IMPROVING U.S. AIRLINE OPERATIONS THROUGH FLIGHT DELAY
ANALYSIS**

PHASE 3

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1.0 Dashboard 1 – Airline Delay Performance

Figure 1 shows the Airline Delay Performance Dashboard developed using Microsoft Power BI. The dashboard was designed to evaluate airline operational performance by analysing flight delays, on-time performance and the primary causes of delays across major U.S. airlines. The dashboard integrates information from the Airline_KPIs and FlightDelay_Master datasets and includes interactive Airline and Season slicers for data exploration.

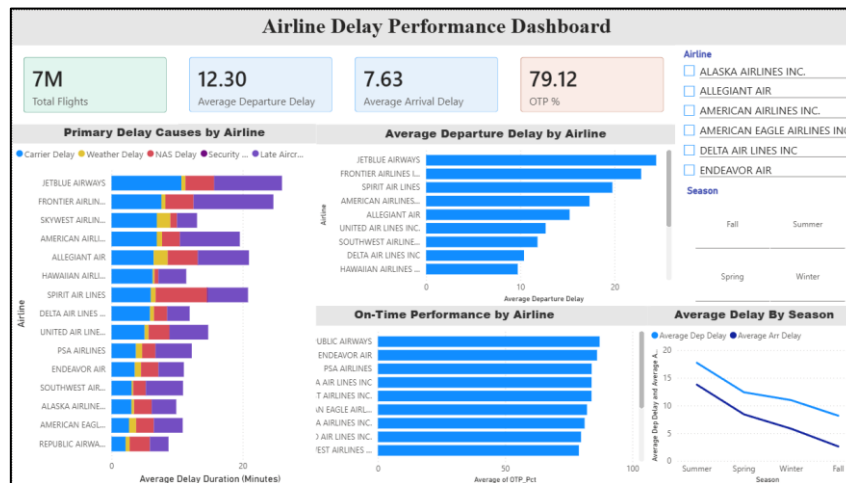


Figure 1: Airline Delay Performance Dashboard

This dashboard is divided into four **Key Performance Indicators (KPIs)**. These KPI cards provide a quick overview on the airline's operational performance and help users to easily evaluate their overall performance.

The **Average Departure Delay by Airline** is a bar chart visualization that used to provide a comparison of the performance of each airline in terms of departure delay. The bar chart was chosen as it can provide an easy comparing view for users to compare the average delay of each airline.

The **Primary Delay Causes by Airline** stacked bar chart was created to examine the impact of various delay categories. This chart can be used to perform a root cause analysis as it provides an understanding of the major reasons for flight disruptions for each airline. A stacked bar chart was chosen as it was the most suitable graph for showing the overall duration of delay and the breakdown of the delay causes.

The **On-Time Performance by Airline** chart was provided to assess airline on-time performance and service reliability. One of the most used performance measures in the aviation industry is On-Time Performance (OTP), which is a crucial measure to assess the operational performance of an airline.

The **Average Delay by Season** line chart was created to analyse the seasonal delay trends by comparing the average departure delay and average arrival delay for Spring, Summer, Fall and Winter. The line chart was chosen because it is best suited to show trends and changes over time.

2.0 Dashboard 2 – Route & Aircraft Performance

Figure 2 shows Route & Aircraft Performance Dashboard developed using Microsoft Power BI. The dashboard was designed to evaluate operational performance at the route level and assess the impact of aircraft age on flight delays across U.S. flight network. The dashboard integrates data from

Route_Delay_Ranking and FlightDelay_Master datasets and includes Season and Aircraft Age Group slicers for data exploration.

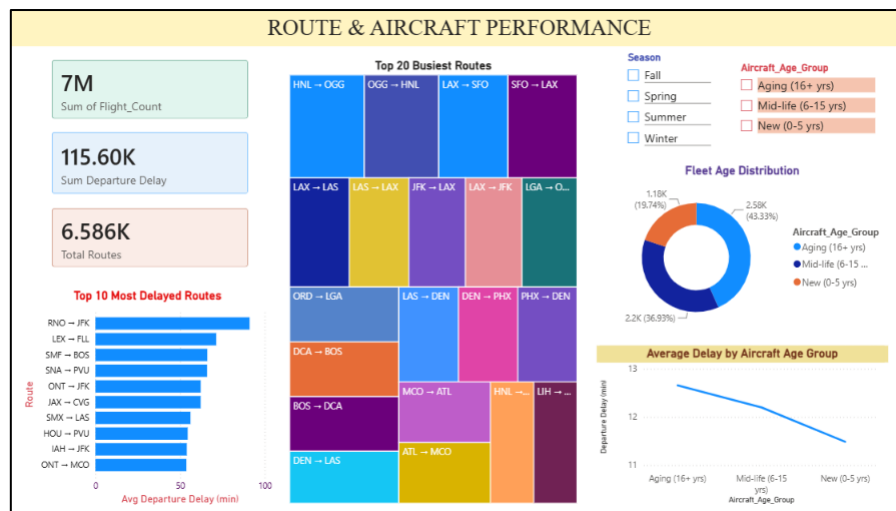


Figure 2: Airline Delay Performance Dashboard

This dashboard is divided into three **Key Performance Indicators (KPIs)**. These KPI cards summarize the total number of flights analyzed, the average departure delay across all routes and the total number of unique routes covered in the analysis. It provides users a quick overview of the scale and overall performance of the route network.

The **Top 10 Most Delayed Routes** is a bar chart visualization used to identify the specific routes with the highest average departure delay. A bar chart was chosen as it allows users to easily compare and rank the routes against one another, making it easy to pinpoint the worst-performing routes that may require operational attention.

The **Top 20 Busiest Routes** treemap illustrates the distribution of flight traffic across major routes. By representing flight volume through proportional sizes, the treemap helps users to quickly identify the routes that handle the largest number of flights.

The **Average Delay by Aircraft Age** line chart shows the average departure delay for different aircraft age categories. For instance, it is classified into three age groups: **New (0–5 years)**, **Mid-life (6–15 years)**, and **Aging (16+ years)**. The chart displays the average departure delay for each group, enabling users to evaluate how delay performance varies with aircraft age and determine whether older aircraft are more prone to operational delays.

The **Fleet Age Distribution** donut chart presents the proportion of aircraft in each age group within the fleet. This provides additional context for understanding the delay patterns observed across different aircraft age categories.

3.0 Storyboard

As can be seen from the storyboard in Figure 3, the major takeaways from the U.S. Airline Delay study have been summarized in one single picture. This storyboard helps managers understand who faces the most delays, which are most serious in which periods, what causes them and their location.

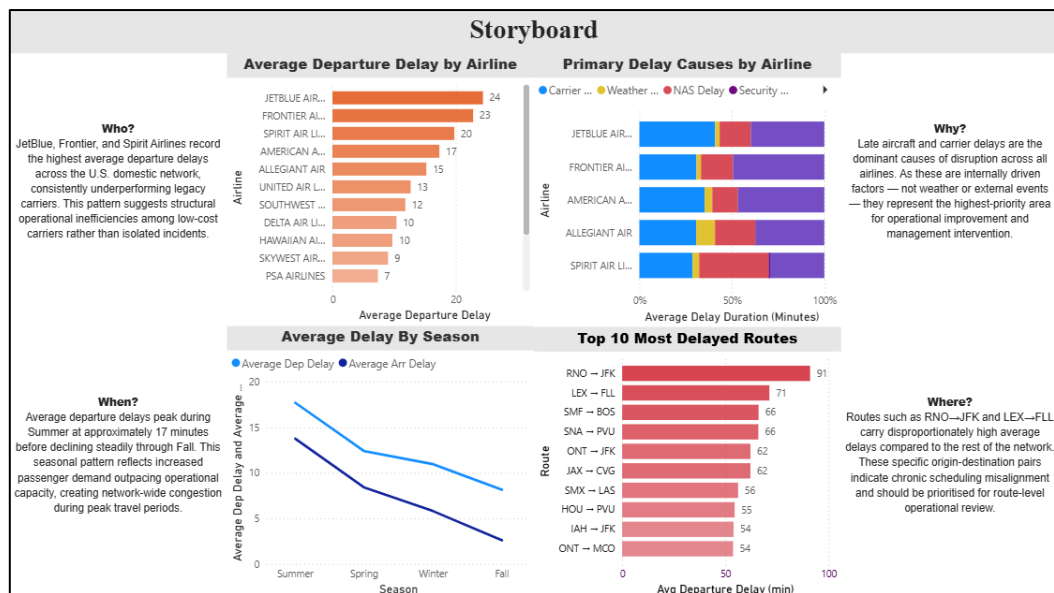


Figure 3: Executive Storyboard

As can be observed from Figure 3, the worst-performing airlines with regard to delays are JetBlue, Frontier and Spirit airlines. The delays are most serious during Summer. Late arrivals and carriers have been identified as the major causes behind this delay. Some of the routes that have faced the highest number of delays include RNO → JFK and LEX → FLL.

4.0 Key Insights

The season that had the highest average delays was Summer (17 minutes), with delays falling gradually up until Fall. The worst performing airline was JetBlue Airways, followed by Frontier and Spirit Airlines, indicating some challenges faced by low-cost airlines.

Late arriving aircraft and the problem within the air carrier were identified as the most common reasons for delays, showing that almost all problems stemmed from interval operations rather than from outside factors. The overall On-Time Performance rate was 79.12% which means about 1 out of 5 flights departed 15+ minutes behind schedule.

RNO-JFK, LEX-FLL routes had much higher delays than on average. Also, older planes (16+ years old) had higher delays and the biggest delays occurred in major airports such as JFK, LAX and ORD.

5.0 Business Recommendations

Poor performing airlines like JetBlue, Frontier and Spirit should carry out operational audits of their operations with a focus on turnaround efficiency, gate handling and scheduling.

There is a need for additional personnel, gates and aircrafts during summer high season when the delay issue will be handled more effectively. Additionally, there is a need to have better aircraft planning since late arriving aircraft is the leading cause of delays within the system.

To ensure better results in the future, there is a need for airlines to speed up the replacement of older aircraft fleets and have more robust maintenance programs for older planes. There should also be capacity planning and scheduling of airline flights especially from congested airports such as JFK, LAX and ORD.